🏢 Payroll Management System:

A full-stack Payroll Management System built using Django (Frontend) and Flask (Backend API) with MySQL as the database.

This system manages:

- Employee Management
- Leave Management
- Attendance Tracking
- Work From Home Requests
- Announcements
- Query & Reply System
- Status Updates

🛠️ Tech Stack:

Layer	Technology Used
Frontend	Django Templates (HTML, CSS, Tailwind)
Backend API	Flask
Database	MySQL
ORM	Raw SQL (Cursor Based)
Email	Flask-Mail
Environment	Python Virtual Environment

📁 Project Structure:

```
Payroll Management/
│
├── flask_app/            # Flask Backend (API)
│   ├── app.py
│   ├── Database_schema.py
│
├── Payroll_Project/      # Django Project (Frontend)
│   ├── app/
│   │   ├── views.py
│   │   ├── urls.py
│   │   ├── templates/
│   │   └── models.py
│
├── requirements.txt
└── README.md
```

How To Run The Project:

1. **Extract the ZIP File:**
   Extract the project folder and open it in VS Code

2. **Create Virtual Environment:**
   Inside the main project folder:
   python -m venv virtual_env (for Mac)
   virtual_env\Scripts\activate (for windows)
   Activate it:
   source virtual_env/bin/activate
3. **Install Dependencies:**
   pip install -r requirements.txt
4. **Setup MySQL Database:**
   1. Open MySQL
   2. Create database:
      CREATE DATABASE payroll_management;
   3. Update database credentials in: flask_app/app.py
      Example:
      DB_HOST = "localhost"
      DB_USER = "root"
      DB_PASSWORD = "yourpassword"
      DB_NAME = “payroll_management"
5. **Run Flask Server (Backend API):**
   Go to flask_app folder:
   cd flask_app
   python app.py
   Flask runs on: http://localhost:5001
   ⚠️ Keep this running.
6. **Run Django Server (Frontend):**
   Open a new terminal:
   source virtual_env/bin/activate
   cd Payroll_Project
   python manage.py migrate
   python manage.py runserver

Django/Project runs on: http://127.0.0.1:8000

For any changes in Project:

1. For DB changes, remember to quit both django and flask or else no change will happen in DB.
2. For accessing session data in Django go to **get_session_user** in views.py
3. For login validation go to **login_required** in views.py

Insertion in Database:

Use the below command

```
insert into employee_complete_details
values('123456','sanjana','sanju@gmail.com','Female','8176894160','2001-07-09','Hyderabad','ECE','employ
ee','Thirumalesh','kavithasree062@gmail.com','2024-02-15','snowhitte','sudhakar@crimsoninnovative.com','2
024-02-15','kurnool','embeded systems','9515693047','sanjana12','sanju','employee');
insert into employee_complete_details
values('145','sanjana','oreekavitharaman27@gmail.com','Female','8176894160','2001-07-09','Hyderabad','EC
E','employee','Thirumalesh','kavithasree062@gmail.com','2024-02-15','snowhitte','sudhakar@crimsoninnovat
ive.com','2024-02-15','kurnool','embeded systems','9515693047','sanjana12','sanju','employee');
insert into employee_complete_details values('201' ,'vamsi', 'vamsit@gmail.com', 'Male', '8008304861',
'2024-01-10', 'ROOM NO.-112 G PULLAREDDY BOYS HOSTEL', 'Research', 'Graduate-Trainee',
'Sudhakar', 'sudha@gamil.com', '2024-03-27', 'Swarnakanth', 'swarnakanth123@gmail.com', '2024-02-23',
'kurnool', 'C, java, python and web development', '8008304861', 'vedh', '2002', 'Employee');
insert into employee_complete_details values('21695a0528','Kavithasree
oree','oreekavitharaman2@gmail.com','Female','9398834607', '2003-07-09', 'Kurnool', 'CSE', 'intern',
'Swarna', 'kavithasree062@gmail.com', '2024-02-15', 'snowhitte', 'sudhakar@crimsoninnovative.com',
'2024-02-15', 'kurnool', 'AI', '9515693047', 'sree', 'oree123', 'employee');
insert into employee_complete_details values('2675', 'pavan', 'pavan@crimsoninnovative.com', 'Male',
'8176894160', '2001-07-09', 'kurnool', 'CSE', 'employee', 'Swarna', 'oreekavitharaman27@gmail.com',
'2024-02-15', 'snowhitte', 'sudhakar@crimsoninnovative.com', '2024-02-15', 'kurnool', 'Pythonfullstack',
'9515693047', 'pavan1', '0000', 'employee');
insert into employee_complete_details values('57', 'Anu snow', 'anusha@crimsoninnovative.com', 'Female',
'9515693047', '2000-07-09', 'hyderabad', 'CSE', 'employee', 'Swarna', 'oreekavitharaman27@gmail.com',
'2024-02-15', 'snowhitte', 'sudhakar@crimsoninnovative.com', '2024-02-15', 'kurnool', 'AI', '9515693047',
'anusree', 'sree123', 'employee');
insert into employee_complete_details values('CIT052', 'Indla sudhakar', 'indlasudhakar132@gmail.com',
'Male', '9703321705', '1999-11-28', 'Tharakaram Nagar', 'CSE', 'cloud application software engineer',
'swarna', 'swarnakanth@crimsoninnovative.com', '2024-02-21', 'srikanth', 'srikanth@gmail.com',
'2022-06-07', 'kurnnol', 'python,java', '98033124244', 'sudhakar', 'sudha123', 'Admin');
```